

Vehicle Augmented Reality Head-up Display information visualization enhancement algorithm and system

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Abstract—In this paper, we propose an augmented reality head-up display (ARHUD) algorithm and system that can enhance the information visualization for drivers. Recently, Head-up Display (HUD) getting popular in the automotive industry. It can serve the car and other information to the driver efficiently. Therefore, the drivers do not have to move the perspectives to the car dashboard. Finally, it can reduce the accident rate effectively. On the other hand, HUD serves only 2D information, and it is challenging to provide suitable 3D information according to road conditions. To overcome this problem, ARHUD has been developed. ARHUD can generate 3D information and can serve adaptive information for drivers by using a car camera. However, the conventional ARHUD defines content color without considering the road condition. It can cause a critical visualization problem. For example, if the information color and the car color are the same in an actual traffic situation, it is not easy to visualize the content information. In addition, to recognize the road situation in real-time, an image processing machine is needed. However, the conventional machine consumes much electricity, and it can cause vehicle discharge. To solve these problems, we propose an effective processing algorithm and system which can define the content color and recognize the object with high speed without consumes high electricity.

Index Terms—Augmented Reality, Vehicle Head Up Display, Visualization

I. INTRODUCTION

Head-up display (HUD) technologies used by pilots make it easier to see the aircraft information at a glance. It can provide various information without disturbing the pilots' sight [1], [2]. As display technologies advance, HUD has been used in various industry fields. Significantly, the car driver must check the information during the driving through the car dashboard, and it can cause accidents. To provide the

information without disturbing the driver's perspective, the car industries install the HUD systems on the car [2]–[4]. Therefore, drivers can recognize the information quickly, and it can reduce the accident rates effectively. However, the conventional HUD only provides 2D information, and it is not easy to provide information in conjunction with actual road conditions. To provide the 3D adaptive information for drivers, ARHUD has been developed [4]–[7]. ARHUD uses the car windshield as a display. It follows Holographic Optical Element (HOE) system based on photopolymer films. The polymer is attached or compressed to the front glass of the vehicle. When it receives the light sources or HUD contents, film works as a lens and provides three-dimensional (3D) information on the road. Therefore, drivers can gain various information about the car and other additional information such as navigation without moving driver perspectives to the dashboard. Therefore, it can reduce the accident rates [8], [9]. In addition, generating 3D information according to the situation of the road can be more effective because it can provide the information with depth and proper location on the windshield screen. However, when the HUD contents color and the car's color are the same, the car color can cover the ARHUD information. Therefore, the driver cannot recognize the information of the ARHUD. To solve this problem, we propose a method that can solve the visualization problems. To generate appropriate color for ARHUD, it detects the object by using YOLOv3, and analyzes the object color accurately by using a processor [10]. Then, it can define the color of the ARHUD content. However, if the processor consumes a lot of electricity, it can cause a lack of battery. Moreover, the processor size should be small. This is because if the size of the machine is large, the vehicle interior design will be restricted. Therefore, we utilize a GPU machine that satisfy

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all the conditions. This paper is organized as follows. We present the principle of the conventional ARHUD and our proposed method in Section 2. In Section 3, we implement the experimental setup and result to show our proposed method performance. Finally, we present the conclusion in Section 4.

II. AUGMENTED REALITY HEAD-UP DISPLAY

A. Conventional method

ARHUD is the display technology that expresses the information on the actual road situation using HOE [6], [7]. The camera records the road situation, makes content for drivers, and projects the information on the polymer film. Finally, the 3D information will be overlaid on the driver's field of view. For example, the ARHUD shows the direction, speed limit, and other information according to the road situation when the driver sets up the destination. Fig. 1 shows the optical setup of the ARHUD.

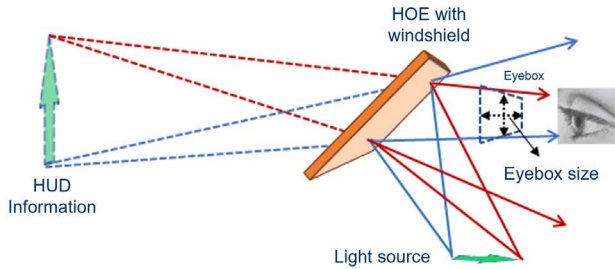


Fig. 1. Augmented Reality Head-Up Display system.

The polymer is attached to the windshield, and the light comes out from the light source such as a projector. The light goes through the polymer film, and ARHUD information is overlaid on the actual road same as fig. 2.

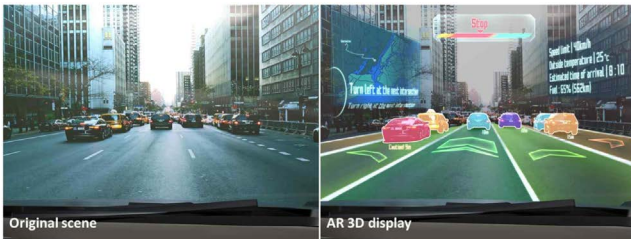


Fig. 2. Augmented Reality Head-Up Display for vehicle.

The conventional HUD can only display the information on limited display space, but the display size can be increased if the polymer film is attached to the whole windshield. Therefore, we can provide not only car information but also other content to drivers. For safety driving, some ARHUD contents contain object recognition algorithm [8], [9]. Through the dashcam video, the processor recognize the car and pedestrians. Then, it project the object information to make the driver aware. Fig. 3 shows examples of the object recognition system.

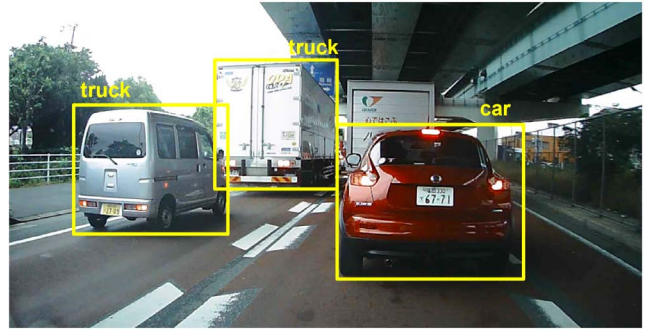


Fig. 3. Utilize object recognition on the real road situation.

This system can reduce accidents and crash with the object in front of the car. However, if the object and ARHUD colors are the same, drivers cannot recognize the information at a glance. If the driver can not recognize the information, it can cause a big accident on the road. Fig. 4 describes the ARHUD visualization problem.



Fig. 4. ARHUD visualization problem.

The word on the white car is readable but the content on the black car is invisible. This is because the car color brightness and the word brightness are the same. Therefore, the color of the contents should be changed according to the object color. In addition, to processing the object recognition system, a high-performance processing machine is needed. However, the size of the processing machine affects the interior design of the car. In addition, if it consumes a lot of electricity, it can cause car discharge. To solve these problems, our proposed method uses a small high-performance processor which consumes a little electricity and can define the HUD content color brightness according to the object color brightness.

B. Proposed method

Our proposed method uses a recorded video frame of the car dashcam as the input image data. The GPU machine detects object and analyzes the color through the input data. Fig. 5 shows our proposed method processing flow chart.

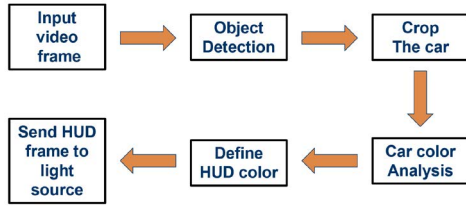


Fig. 5. Proposed method processing flow chart.

To detect the object in the image, we use the YOLOv3. Fig. 6 shows the YOLOv3 network.

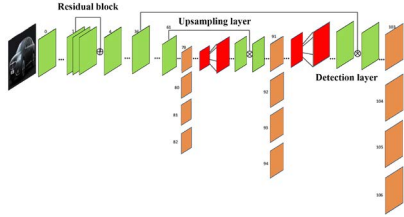


Fig. 6. Object detection YOLOv3 network architecture.

YOLOv3 is the pre-trained object detection algorithm which can detect the object fast and accurately. Our proposed method utilizes YOLOv3 algorithm for object detection. Thus, the drivers can get information about pedestrians and cars in front of the car. Moreover, it provides the location of the object in the scene. Our proposed method crops the object in the image through the location information to analyze the object color as follows.

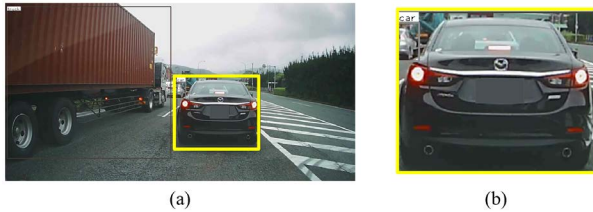


Fig. 7. Object detection image, (a) object detected scene, (b) cropped object image.

Through the cropped image, we can get the RGB value of the image. However, to analyze the color characteristics in detail, our proposed method changes the RGB to the HSV using Eqs. (1), (2) and (3). In addition, fig. 8 shows the difference between RGB and HSV.

$$V = \max(R, G, B) \quad (1)$$

$$S = \begin{cases} \frac{V - \min(R, G, B)}{V} & \text{if } V \neq 0 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

$$H = \begin{cases} 60(G - B) / (V - \min(R, G, B)) & \text{if } V = R \\ 120 + \frac{60 - (B - R)}{V - \min(R, G, B)} & \text{if } V = G \\ 240 + \frac{60 - (R - G)}{V - \min(R, G, B)} & \text{if } V = B \\ 0 & \text{if } R = G = B \end{cases} \quad (3)$$

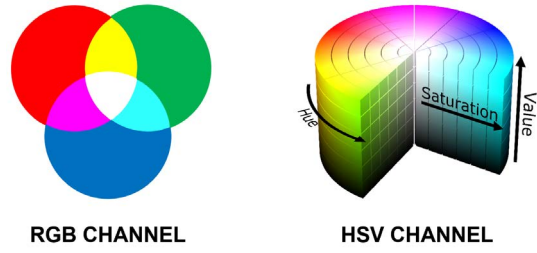


Fig. 8. Difference between RGB and HSV channel.

R, G, B are the values of each channel, respectively. Our proposed method utilizes H (HUE) channel, which represents the car's color. The cropped image contains not only the car color also the background color of the road. Therefore, through the histogram of the H channel and get the median value of the histogram. Then, we make a mask image to analyze the car color more accurately. The exact car S (saturation) channel and V (brightness) channel values can be extracted to multiply with the original and mask images. Finally, the car color can be analyzed into HSV values, respectively.

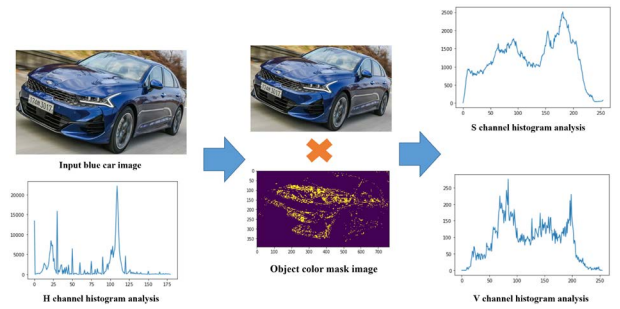


Fig. 9. Object color analysis.

However, if the white or black color car is detected, the H channel cannot express the color precisely. However, if the white or black car is detected, the H channel cannot precisely express the color. This is because the H channel index can not express the white and black colors. Thus, it shows the 0 or low value in the H channel when detecting the black or white object in the scene. To distinguish the black and white color, our proposed method utilizes the S channel and V channel. The saturation median value must be low because of the white color characteristic. Finally, it can define the white car by using Eq. (4).

$$H = 0, S = 0, V = 255 \quad \text{if } H < 10 \text{ and } S < 20 \quad (4)$$

Since the cropped image contains the background and reflection color from the light, it is difficult to detect the utterly white color. To distinguish the white color, the threshold values have been used in H and S channels. If the saturation value is lower than 20, the object recognizes as white color. However, the saturation value is over 20, follows the cropped image brightness value. Therefore, our proposed method can

distinguish the object colors in various situation. In addition, if the car has a color without black and white, we can analyze the car saturation and brightness value more accurately by using the mask image. According to the HSV channel object color analysis result, the customized colormap for the ARHUD must be defined below.

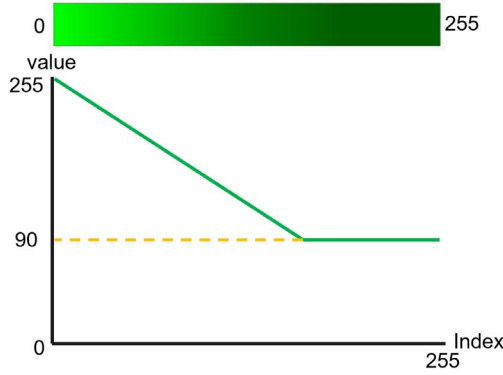


Fig. 10. Custom colormap for ARHUD content.

Fig. 10 shows the color of ARHUD content. Our proposed method utilizes the green color to customize the colormap. This is because the green color is suitable for displaying information without interfering driver's perspective and relieves eye fatigue. In addition, the brightness of HUD content can be adjusted according to the car brightness value and polymer film reflection ratio through the customized colormap. Moreover, to show the contents information only, the HUD display screen is needed. The background must be black and put the HUD contents only according to the object location of the front view as below.

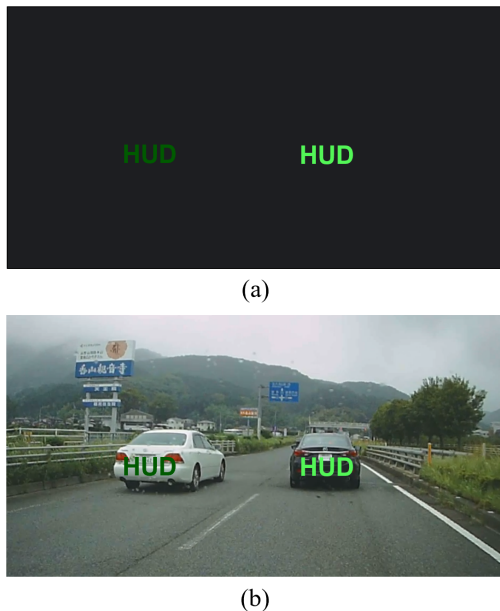


Fig. 11. ARHUD image, (a) ARHUD display, (b) overlaid the contents on the road.

Finally, the ARHUD contents are overlaid on the road, and it can define the content color that is helpful to the driver to recognize the information. In Section 3, we implement the optical simulation experiment, and compare the conventional result with the proposed result to show visualization differences.

III. EXPERIMENTAL SETUP AND RESULT

In our experiment, we use the Jetson Xavier to processing the image through the YOLOv3. It consumes low electricity, and it can process many images by using GPU. The device size is also small, so it can be suitable for processing the vehicle ARHUD contents. It saves approximately seven times as much electricity as a GPU mounted on a typical desktop. To simulate the actual road situation, we use a recorded dashcam video which has 1280×720 resolution. The dashcam video frame operates as input data, and the GPU machine processes the object detection, color analysis and generates the HUD contents display. Fig. 12 shows our simulation flow chart.

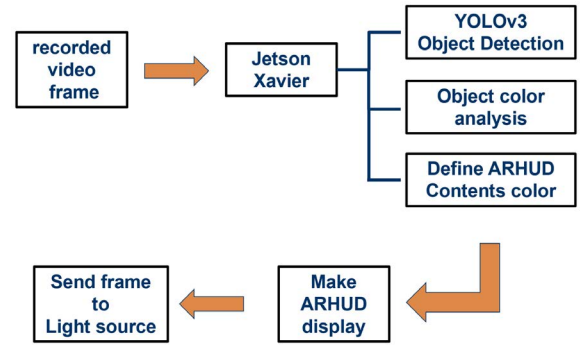


Fig. 12. Simulation experimental setup.

To provide the object information to driver, we make contents as below fig. 13.

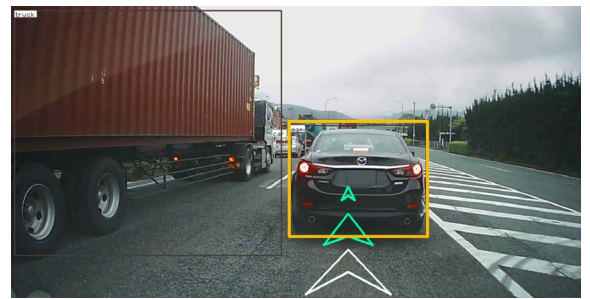


Fig. 13. ARHUD contents.

If the object position overlaps with the HUD contents, the color of contents can be defined according to the detected object color. If the object is not overlapped with the content, we defined the content color is white. To demonstrate the color definition performance, we present the proposed method results when a white car is recognized from the driver's perspective.



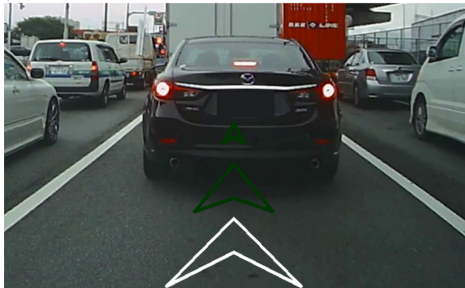
(a)



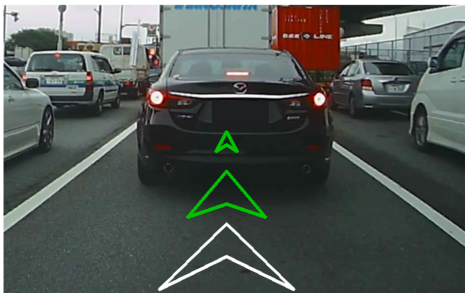
(b)

Fig. 14. white car result, (a) conventional method, (b) proposed method.

Fig. 14 shows the ARHUD result images. The processor recognizes the car as a white color and defines the low brightness color according to the customized colormap for ARHUD content. Therefore, it can enhance the ARHUD content visualization. If content shows a bright color same as the object color, it can cause significant irritation to the eyes. Fig. 15 and 16 shows the result images when our proposed method recognizes the various car color in the scene.

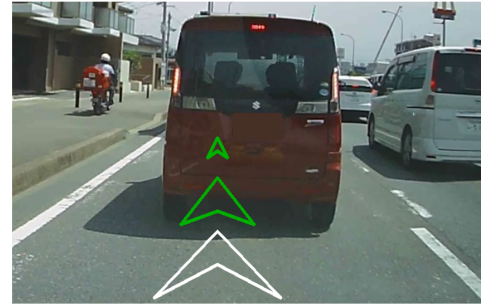


(a)

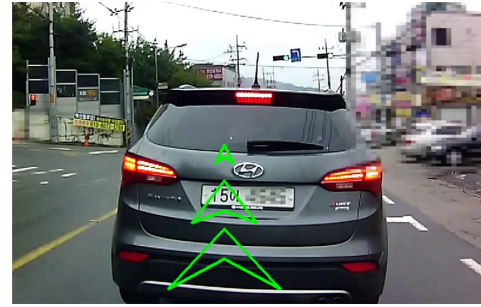


(b)

Fig. 15. Black car result, (a) conventional method, (b) proposed method.



(a)



(b)

Fig. 16. Diverse car color result of proposed method, (a) red car, (b) dark blue car.

Finally, our proposed method can define the ARHUD content color according to the object color overall. The ARHUD content of the proposed method is recognizable. However, the conventional method may not provide the proper color of the content. Even the car has various colors, our proposed method can generate appropriate content color because it utilizes the mask image to get the accurate brightness value of the car. Even if the color of the vehicle changes according to reflected light and weather, it can handle the various situations by analyzing the object color through mask image analysis.

IV. CONCLUSIONS

In this paper, we proposed an algorithm using a GPU machine which can analyze the object color and define the ARHUD content color according to the vehicle color. Our research focused on the efficiency of algorithm and system. Especially, to consider the diversity of vehicle interior design and space in the vehicle, the size of the processing machine must be small, and power consumption must be small because of the battery discharge. Therefore, we used a suitable GPU machine called Jetson Xavier. To recognize the object in the scene using a GPU machine, we used YOLOv3, which can recognize the car and pedestrian in the scene. In addition, it can define the proper color for the ARHUD content color according to the object color. Finally, our proposed method can enhance the visualization of the ARHUD. However, if the camera has 1280×720 resolution, it can generate 14 to 17 frames per second. It is insufficient to generate content in

real-time, but parallel processing optimization can solve this problem. We will find out the solutions in the future.

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